Organisation of the organism

Mark scheme and worked solutions for 20 questions: 10 multiple choice, 10 structured.

1 · Part 1

Core · Cell structure · [1]

Answer: B

Explanation

Plant cells have a cell wall (made of cellulose) outside the cell membrane; animal cells do not. The cell membrane, mitochondria and ribosomes are found in both plant and animal cells, so options A, C and D are wrong.

2 · Part 1

Core · Cell structure · [1]

Answer: D

Explanation

Bacterial cells have a cell wall and plasmids (small rings of DNA), but they have no nucleus: their main DNA is a single circular loop in the cytoplasm. Row A wrongly includes a nucleus and leaves out plasmids, row B leaves out the cell wall, and row C leaves out the cell wall and includes a nucleus.

3 · Part 1

Core · Cell structure · [1]

Answer: C

Explanation

Ribosomes are the site of protein synthesis. Aerobic respiration happens in mitochondria, the cell membrane controls which substances enter and leave, and cell sap is stored in the vacuole, so options A, B and D describe other structures.

4 · Part 1

Core · Cell structure · [1]

Answer: A

Explanation

Mitochondria are the site of aerobic respiration, which releases energy. Wing muscle cells contract a lot and need much more energy than fat-storage cells, so they have more mitochondria. Protein synthesis happens at ribosomes, genetic material is in the nucleus, and the cell membrane controls entry of substances, so options B, C and D link mitochondria to the wrong function.

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5 · Part 1 Core Cell structure [1]

Answer: D

Explanation

Ciliated cells move mucus in the trachea and bronchi. Neurones conduct electrical impulses, root hair cells are for absorption, and red blood cells transport oxygen, so the pairings in options A, B and C are all mixed up.

6 · Part 1 Core Cell structure [1]

Answer: B

Explanation

Muscle tissue is a group of similar cells working together, the stomach is an organ made of several tissues, and the digestive system is an organ system made of several organs. The other rows place at least one of these at the wrong level of organisation.

7 · Part 1 Core Cell structure [1]

Answer: A

Explanation

New cells are always produced by the division of existing cells. Bacterial cells and red blood cells have no nucleus, so option B is wrong; animal cells also contain mitochondria, so option C is wrong; and animal cells have no cell wall, so option D is wrong.

8 · Part 1 Core Size of specimens [1]

Answer: C

Worked solution

Both lengths are already in millimetres, so divide: 36 ÷ 0.45 = 80. Check: 0.45 mm × 80 = 36 mm, which matches the drawing.

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9 · Part 1 Core Size of specimens [1]

Answer: A

Worked solution

Rearrange magnification = image size ÷ actual size to give actual size = image size ÷ magnification. 15 mm ÷ 250 = 0.06 mm. Check: 0.06 mm × 250 = 15 mm.

10 · Part 1 Extended Size of specimens [1]

Answer: D

Worked solution

There are 1000 µm in 1 mm, so the image size is 40 000 µm. magnification = image size ÷ actual size = 40 000 µm ÷ 5 µm = 8000. Check another way: 5 µm = 0.005 mm, and 40 mm ÷ 0.005 mm = 8000.

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11 · Part 1 Core Cell structure [5]

Mark scheme

(a) **[2]**

- any two from: cell membrane / nucleus / cytoplasm / mitochondria / ribosomes (1) (1)

IGNORE vacuole

REJECT cell wall, chloroplasts

(b) **[2]**

- cell wall (1)
- chloroplast(s) (1)

(c) **[1]**

- the plant cell has a large (permanent) vacuole (containing cell sap), while the liver cell has only small (temporary) vacuoles (1)

ALLOW the plant cell vacuole is much larger / permanent

Worked solution

(a) Plant and animal cells share the basic parts of a cell: a cell membrane around the outside, cytoplasm inside it, a nucleus containing the genetic material, mitochondria for aerobic respiration and ribosomes for making proteins. Any two of these earn the marks.

(b) A pondweed leaf cell photosynthesises, so it contains chloroplasts. Like all plant cells it also has a cell wall outside the cell membrane. Liver cells have neither.

(c) Plant cells have one large, permanent vacuole filled with cell sap. Animal cells such as liver cells have no large vacuole; any vacuoles they have are small and temporary. The key word for the mark is the size (large compared with small) or permanence of the vacuole.

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12 · Part 1 Core Cell structure [6]

Mark scheme

(a) **[2]**

- (one) circular DNA / a loop of DNA (in the cytoplasm) (1)
- plasmids / small rings of DNA (1)

(b) **[2]**

- any two from: cell membrane / cytoplasm / ribosomes (1) (1)

REJECT cell wall, nucleus, mitochondria

(c) **[1]**

- mitochondria / mitochondrion (1)

ALLOW (small / temporary) vacuole(s)

(d) **[1]**

- cell wall (1)

REJECT plasmids, circular DNA (excluded by the question)

Worked solution

(a) A bacterial cell has no nucleus. Its main DNA is a single circular loop lying in the cytoplasm, and it also has plasmids, which are small extra rings of DNA. One mark is for each form.

(b) Bacterial cells and human cells both have a cell membrane, cytoplasm and ribosomes. A cell wall is found in bacteria but not in human cells, so it does not count.

(c) Human cells have mitochondria for aerobic respiration; bacterial cells do not have mitochondria. The question has already ruled out the nucleus.

(d) Bacterial cells have a cell wall outside the cell membrane. Human skin cells, like all animal cells, have no cell wall. Plasmids and circular DNA are excluded by the question.

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13 · Part 1 Core Cell structure [6]

Mark scheme

(a) **[2]**

Any two from:

- the cell cannot control which substances enter and leave (1)
- useful substances / cell contents leak out of the cell (1)
- harmful or unwanted substances can enter the cell (1)
- the cell may die (1)

(b) **[1]**

- cytoplasm (1)

(c) **[2]**

- contains the genetic material / DNA / chromosomes (1)
- controls the activities of the cell (1)

(d) **[1]**

- stores / contains cell sap (1)

ALLOW keeps the cell firm / turgid / supports the cell

Worked solution

(a) The cell membrane controls which substances enter and leave the cell. If it is damaged, the cell loses this control: useful substances such as glucose or proteins can leak out, and harmful substances can get in. Without this control the cell may die. Any two sensible effects earn the marks.

(b) The cytoplasm is the jelly-like part of the cell where most chemical reactions happen.

(c) The nucleus contains the genetic material (DNA in chromosomes). The genetic material controls what the cell does, so one mark is for what the nucleus contains and one mark is for its control of the cell's activities.

(d) The large permanent vacuole of a plant cell holds cell sap, a solution of water with dissolved sugars and ions. When it is full it presses outwards and keeps the cell firm.

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14 · Part 1 Core Cell structure [7]

Mark scheme

(a) **[2]**

- P ciliated cell; Q neurone; R palisade mesophyll cell; S root hair cell: all four correct (2), two or three correct (1)

ALLOW nerve cell for neurone; palisade cell for palisade mesophyll cell

(b) **[1]**

- conduction of electrical impulses (1)

ALLOW carries nerve impulses / electrical signals

(c) **[2]**

- (the extension) increases the surface area (of the cell) (1)
- so more water / mineral ions can be absorbed (from the soil) (1)

ALLOW the extension reaches more of the soil for absorption

(d) **[2]**

- gametes (1)
- reproduction (1)

Worked solution

(a) Use the clues. Hair-like structures lining the trachea are cilia, so P is a ciliated cell. A cell body with a very long fibre running from the spinal cord to the foot is a neurone (Q). A column-shaped cell packed with chloroplasts just below the upper surface of a leaf is a palisade mesophyll cell (R). A cell on the outside of a root with an extension between the soil particles is a root hair cell (S).

(b) Neurones conduct electrical impulses around the body.

(c) The job of a root hair cell is absorption. The long, thin extension gives the cell a much larger surface area in contact with the soil and the water around soil particles, so more water and mineral ions can be absorbed. One mark is for the larger surface area and one for linking it to more absorption.

(d) Sperm and egg cells are gametes, and their function is reproduction.

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15 · Part 1 Core Cell structure [7]

Mark scheme

(a) **[2]**

- a group of cells with similar structures (1)
- working together to perform a shared function (1)

(b) **[2]**

- a structure made up of a group of tissues (1)
- working together to perform specific functions (1)

REJECT group of cells for the first mark

(c) **[2]**

- organ: stomach (1)
- organ system: digestive system (1)

(d) **[1]**

- any one from: leaf / root / stem / flower (1)

Worked solution

(a) A tissue is a group of cells with similar structures that work together to carry out a shared function, for example muscle tissue contracting.

(b) An organ is a structure made of several different tissues working together to perform specific functions. The stomach contains muscle tissue and other tissues that together digest food.

(c) The stomach is an organ. The digestive system is a group of organs, including the stomach, that work together, so it is an organ system. The red blood cell is a single cell and muscle tissue is a tissue.

(d) Plant organs include the leaf, stem, root and flower. Each contains several tissues working together.

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16 · Part 1 Core Cell structure [6]

Mark scheme

(a) **[3]**

- Z (1)

Any two reasons from:

- has no nucleus (1)
- has plasmids (1)
- has no mitochondria (1)

ALLOW has a cell wall but no nucleus as one reason

(b) **[2]**

- root / any part that is underground or not exposed to light, e.g. inside of a stem (1)
- there is no light there, so the cell cannot photosynthesise / chloroplasts would have no use (1)

(c) **[1]**

- it has no cell wall (1)

ALLOW it has no large permanent vacuole

Worked solution

(a) Cell Z is the bacterium. Bacterial cells have no nucleus (their DNA is a circular loop in the cytoplasm), they have plasmids, and they have no mitochondria. Cell Z is the only cell with all of these features, and it also has a cell wall, which bacteria have. Give any two of these features from the table as your reasons.

(b) Chloroplasts are used for photosynthesis, which needs light. Cells in the roots grow underground where no light reaches them, so they cannot photosynthesise and have no chloroplasts. One mark is for a sensible part of the plant without light, such as the root, and one mark for the reason.

(c) Cells X and Y both have a nucleus and mitochondria, so they are the plant and animal cells. Cell Y has no cell wall, and animal cells never have a cell wall, so Y is the animal cell. Cell Y also lacks the large permanent vacuole that plant cells have.

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17 · Part 1 Core Size of specimens [4]

Mark scheme

(a) **[1]**

- magnification = image size ÷ actual size (1)

ALLOW equivalent rearrangement or words, e.g. size of drawing ÷ real size

(b) **[2]**

- 60 ÷ 0.12 (1)
- × 500 (1)

ALLOW correct answer with no working for both marks

IGNORE units given with the magnification

(c) **[1]**

- the magnification increases / is greater (1)

Worked solution

(a) magnification = image size ÷ actual size.

(b) Both measurements are in millimetres, so divide the drawing size by the real size: 60 mm ÷ 0.12 mm = 500. The magnification is × 500. Check: 0.12 mm × 500 = 60 mm.

(c) The actual size of the egg cell does not change. If the image is bigger, image size ÷ actual size gives a bigger number, so the magnification of the second drawing is greater.

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18 · Part 1 Core Size of specimens [6]

Mark scheme

(a) **[2]**

- 50 ÷ 20 (1)
- 2.5 mm (1)

(b) **[2]**

- 36 ÷ 0.4 (1)
- × 90 (1)

(c) **[2]**

- 3.2 × 15 (1)
- 48 mm (1)

ALLOW , in each part, a correct answer with no working for both marks

Worked solution

Start from magnification = image size ÷ actual size and rearrange it for what you need.

(a) actual size = image size ÷ magnification = 50 mm ÷ 20 = 2.5 mm. Check: 2.5 mm × 20 = 50 mm.

(b) magnification = image size ÷ actual size = 36 mm ÷ 0.4 mm = 90, so the magnification is × 90. Check: 0.4 mm × 90 = 36 mm.

(c) image size = actual size × magnification = 3.2 mm × 15 = 48 mm. Check: 48 mm ÷ 15 = 3.2 mm.

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19 · Part 1 Extended Size of specimens [6]

Mark scheme

(a) **[2]**

- 75 ÷ 25 000 (1)
- 0.003 mm (1)

ALLOW 3×10^{-3} mm

(b) **[1]**

- 3 µm (1)

ECF from (a): answer to (a) × 1000

(c) **[2]**

- 0.5 ÷ 25 000 = 0.00002 mm (1)
- 0.02 µm (1)

ALLOW 2×10^{-5} mm for the first mark; 2×10^{-2} µm for the second mark

(d) **[1]**

- 150 (times) (1)

ALLOW ECF from (b) and (c)

Worked solution

(a) actual size = image size ÷ magnification = 75 mm ÷ 25 000 = 0.003 mm.

(b) There are 1000 µm in 1 mm, so 0.003 mm × 1000 = 3 µm.

(c) The ribosome: 0.5 mm ÷ 25 000 = 0.00002 mm. Multiply by 1000 to change to micrometres: 0.00002 × 1000 = 0.02 µm.

(d) Divide the length of the mitochondrion by the diameter of the ribosome, in the same units: 3 µm ÷ 0.02 µm = 150. Check a second way: both are in the same photograph, so the ratio of their image sizes is the same, 75 mm ÷ 0.5 mm = 150.

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20 · Part 1 Extended Size of specimens [6]

Mark scheme

(a) **[1]**

- 0.0075 mm (1)

ALLOW 7.5×10^{-3} mm

(b) **[2]**

- 30 ÷ 0.0075 (1)
- × 4000 (1)

ALLOW ECF from (a)

ALLOW 30 000 µm ÷ 7.5 µm for the first mark

(c) **[1]**

- did not convert the two sizes into the same units / divided 30 mm by 7.5 µm directly (1)

(d) **[2]**

- 6 ÷ 4000 = 0.0015 mm (1)
- 1.5 µm (1)

ALLOW ECF from (b)

Worked solution

(a) There are 1000 µm in 1 mm, so divide by 1000: 7.5 µm ÷ 1000 = 0.0075 mm.

(b) Use the same units for both sizes: magnification = 30 mm ÷ 0.0075 mm = 4000, so the magnification is × 4000. Check in micrometres: 30 mm = 30 000 µm, and 30 000 µm ÷ 7.5 µm = 4000.

(c) 30 ÷ 7.5 = 4. The second student divided millimetres by micrometres without first converting them to the same unit, so the answer is 1000 times too small.

(d) actual size = image size ÷ magnification = 6 mm ÷ 4000 = 0.0015 mm. Convert to micrometres: 0.0015 × 1000 = 1.5 µm. Check: 1.5 µm × 4000 = 6000 µm = 6 mm.

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